

# SVG → React-PDF Converter: Architecture Design

This document defines the exact technical architecture for a tool that converts standard SVG files into React-PDF compatible SVG JSX.

## 1. System Overview

The system is a compiler-style pipeline that transforms SVG input into React-PDF JSX output. Each stage is isolated, testable, and replaceable.

Pipeline Flow:

- Input Loader (SVG file or string)
- SVG Parser (XML → AST)
- Normalizer (cleanup & standardization)
- Compiler (SVG AST → React-PDF AST)
- Validator (unsupported feature detection)
- Emitter (JSX + imports output)

## 2. Core Modules

- **Parser:** Uses `svgson` to convert SVG XML into a JSON AST.
- **Normalizer:** Converts styles to attributes, camelCases props, ensures `viewBox`, removes noise.
- **Compiler:** Maps SVG tags to React-PDF components (`svg` → `Svg`, `path` → `Path`, etc.).
- **Validator:** Detects unsupported SVG features and emits warnings instead of failing.
- **Emitter:** Generates clean JSX and required imports from `@react-pdf/renderer`.

## 3. Data Structures

**SVG AST Node:** name, attributes, children

**React-PDF AST Node:** component, props, children

## 4. Supported Scope (v0.1)

- `svg`, `path`, `rect`, `circle`, `line`, `polyline`, `polygon`, `g`
- `fill`, `stroke`, `strokeWidth`, `opacity`
- No filters, masks, gradients, or animations

## 5. Output

The final output is valid React-PDF JSX with auto-generated imports. The result can be pasted directly into React-PDF components.

## 6. Extension Roadmap

- Transform flattening
- Gradient support
- Web-based SVG upload UI
- VS Code extension
- Figma export plugin